A defining polynomial in twenty-seven variables: an existence theorem with a uniform degree bound

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Abstract

Under the four arithmetic inputs of the specified elliptic source, we prove that the integers are the rational input projection of an integer polynomial in twenty-seven total variables and of degree at most 2^{158}. We give a computable family F_L of such polynomial specifications, indexed by a fixed odd integer coefficient parameter L. Every sufficiently large L has the required input projection. The proof uses von Buhren's uniform Mordell–Lang height theorem; its ineffective additive constant prevents us from certifying a numerical sufficient value of L. Thus the variable and degree bounds are proved, while no particular member is asserted here to be a fully explicit defining polynomial. Exact source, finite-algebra, and degree certificates accompany the family.

1 Statement and arithmetic inputs

All variable counts include the distinguished input. A polynomial $F \in \mathbb{Z}[X, Y_1, \dots, Y_m]$ defines $\mathbb{Z}$ over $\mathbb{Q}$ if

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^m \quad F(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

The integer L below is fixed when a polynomial is constructed; it is neither its input nor an existential variable.

Theorem 1. *Assume the elliptic group and height assertion, the exponent nineteen contact-height bound, the zero-preparation pole completeness assertion, and the simultaneous prime-pattern input specified in Section 1.1. There is a computable family*

$$F_L \in \mathbb{Z}[X, Y_1, \dots, Y_{26}], \qquad L \geq 3 \text{ odd},$$

of finite polynomial specifications with the following properties.

1. *Every member has degree at most 2^{158}.*
2. *There is a fixed real number L_0 such that every odd $L \geq L_0$ defines $\mathbb{Z}$ over $\mathbb{Q}$.*
3. *The proof does not determine an effective sufficient value of L_0, and does not give an effective coefficient-height bound for a member known to define $\mathbb{Z}$.*

The finite specification uses integer coefficient arithmetic, determinants, resultants, traces of finite matrices, and finite products, all with explicit finite index ranges. It uniquely specifies an ordinary polynomial, without treating the matrix coordinates or algebra generators as quantified variables.

1.1 The four inherited inputs

Set

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \quad E : y^2 = x^3 - Dx,$$
$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

The construction uses the following assertions, with exactly the scope established in the arithmetic dependency paper [3].

1. $2E(K) = \mathbb{Z}P$, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.
2. For the five normalized contacts with $\infty, 0, 1, 2401, -343/32$, the odd contact radical $M_5(q)$ outside primes at most 76176 satisfies $h(q) \leq HM_5(q)^{19}$ when q is not a contact value, where
$$H = 2^{110}210^{1800}\left(\prod_{p \leq 76176} p\right)^{19} + 128.$$
3. With $B = 17b$, the pole system with preparation parameter zero is
$$p^2 = B(1-e)(1+3e), \qquad Bq^2 = (s^2 - Be(B-b^2))(s^2 + 3Be(B-b^2)).$$
Its solvability excludes negative odd $v_\ell(b)$ for $\ell \notin \{2, 3, 17\}$. It is soluble whenever $0 < b < 17$, its reduced denominator is square, it is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and its valuations there are at least $3, 1, 2$, respectively.
4. The simultaneous prime-pattern input permits the source's prescribed prime values, specified Gaussian generators, and reduction-order conditions, arbitrary fixed divisibility, and arbitrary fixed odd avoidance. It applies to the finite collection of comparison scales after the root points have been fixed.

The prime-pattern input includes the stated application of Sun's theorem. No new conjecture is introduced by the height theorem used below: von Buhren's result [7] is a published theorem. Its effect here is ineffectivity of one fixed coefficient.

2 The sum of four traces controls the four labels

For nonzero integers $n_1, \ldots, n_4$ put

$$T(\mathbf{n}) = \mathrm{Tr}_{K/\mathbb{Q}}\left(\sum_{i=1}^{4} x([n_i]P)\right).$$

The aggregate trace-height theorem [1] proves that there are fixed constants $C > 0$ and $J_0 \geq 0$ such that

$$\max_i |n_i|^2 \leq Ch(T(\mathbf{n})) + J_0. \tag{1}$$

The multiplicative constant comes from fixed geometric data; the additive constant is ineffective.

For clarity, the essential hypotheses of this application are as follows. After rescaling the two conjugate curves to $E_0 : y^2 = x^3 - x$, their source points are independent over $\mathbb{Z}[i]$. The sum of their abscissas has no constant restriction to an abelian translate through a tuple of nonzero multiples. Within one block, coincident poles have tangent multipliers in the rational ratios of their integer labels, so the leading coefficient is a nonzero sum of positive rational squares. Poles in different blocks cannot coincide by independence. The closed graph of the trace function therefore avoids the exceptional abelian translates at every source tuple. Applying von Buhren's theorem to this fixed graph and its fixed finite-rank subgroup gives (1), including a uniform additive constant. The companion paper [1] supplies the full proof over the algebraic closure, so translates defined only over an extension field are included.

3 The exact rational source

We use the point charts, local tests, and denominator guards of the established source [2]. We describe their data and the changes defining the present family.

There are nine points over K, named

$$A, S, T_1, Z, H_0, U_A, U_S, U_T, U_Z.$$

For each point retain two rational coordinates h_0, h_1 and set

$$h = h_0 + h_1\sqrt{2}, \qquad x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj. \tag{2}$$

This chart describes precisely the nonzero points of $\mathbb{Z}P$. Indeed, let $R_0 = (1+\sqrt{2}, 2\sqrt{2})$, so $2R_0 = P$. The group input implies that every $Q \in E(K)$ is $mR_0 + T$ with $T \in E[2](K)$. For the standard two-isogeny descent map, $\delta(R_0) = 1+\sqrt{2}$ has opposite signs at the real embeddings, whereas the torsion classes are $1, -D, d, -d$, with equal signs. A square abscissa therefore forces m even. Of these torsion classes only 1 is square: $-D, -d$ are negative and $N_{K/\mathbb{Q}}(d) = 7$ is not a rational square. Thus $Q \in \mathbb{Z}P$. Conversely, $\delta(2R) = 1$ gives a square abscissa for every nonzero multiple of P. Neither $h = 0$ nor $j = 0$ is possible in (2). A direct proof of the descent map and all these assertions is given in [5].

The finite root j is represented by one monic quartic, without a finite choice of charts. Write

$$h^4 - D = A + B\sqrt{2}, \quad A = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \quad B = 4h_0h_1(h_0^2 + 2h_1^2) + 6.$$

Set

$$c = 2A + B, \qquad d_1 = -A - B, \qquad j' = (c + d_1\sqrt{2})j.$$

Writing $(c + d_1\sqrt{2})^2(A + B\sqrt{2}) = A' + B'\sqrt{2}$ gives

$$A' = 6A^3 - 9AB^2 - 4B^3, \qquad B' = -4A^3 + 6AB^2 + 3B^3.$$

For $j = u + v\sqrt{2}$, direct expansion gives

$$j' = 2(u^3 - 2v^3) + (4v^3 - u^3)\sqrt{2}.$$

Its rational component is nonzero, since 2 is not a rational cube and $j \neq 0$. The multiplier has norm $2A^2 - B^2$, which is nonzero for rational $(A, B) \neq (0, 0)$ because 2 is not a rational square. Thus the reconstruction uses only these two elementary rational obstructions. The general quadratic extension version is proved in [6]. We may therefore use the single chart

$$Y^4 - 4A'Y^2 + 8(B')^2 = 0, \quad Y \neq 0, \qquad j' = Y/2 + (B'/Y)\sqrt{2}, \qquad j = \frac{j'}{c + d_1\sqrt{2}}. \tag{3}$$

The source retains the nonzero quartic derivative, the norm of j', and every denominator of the affine point operations. The root is simple: a vanishing derivative at a nonzero rational root would force the two rational components of j' to have squares in the ratio 2, and hence both to vanish.

The four root labels (n, a, t, b) are nonzero integers; no parity condition is needed for soundness. The simultaneous completeness construction retains its original odd labels. The scalar shift is now computed directly as $S - P$, in the affine difference chart. Its excluded equal-abscissa cases are $S = \pm P$, both avoided by that construction.

Put $Q_0 = P$, $(Q_1, Q_2, Q_3, Q_4) = (A, S, T_1, Z)$, $U_0 = H_0$, and $V_i = U_i - Q_i$. Write $z = -x/y$ and $f = 1/x$. The truncated formal logarithms are

$$\ell_r(Q) = z(Q)\sum_{j=0}^{r/4-1} c_j(Df(Q)^2)^j, \quad c_0 = 1, \quad c_1 = -2/5, \quad c_j = c_{j-1}\frac{4j-5}{4j+1}.$$

Use precisions $(12, 40, 40, 40)$, ratio orders $(10, 39, 39, 39)$, and scalar order 20. With $\beta = 216X^4 + 450459$, the sixteen tests are the ten formal-group tests $f(U_i), \sigma f(V_i)$, the four differences between the paired logarithmic ratios, the direct target test for β, and the direct Lucas polynomial

$$v^2 - u(u-2)vw + (u-2)^2w^2 - 1$$

in the three corresponding ratios. Their assigned orders are 1 for each of the first ten tests and $10, 39, 39, 39, 10, 39$ for the remaining six. These tests and their construction are unchanged.

3.1 Two comparison scales

We use precisely the two positive rational square scales

$$A_0 = 1, \qquad A_1 = \left(\mathrm{Tr}_{K/\mathbb{Q}} \sum_{i=1}^{4} x(Q_i) \right)^2. \tag{4}$$

They are nonzero: the root points are nonzero doubles and their abscissas are positive in both real embeddings. For the retained slope λ, each scale is inserted into the same normalized comparison detector of [2]:

$$t_r = A_r\lambda, \quad \delta_r = t_r - 1, \quad F_r = \prod_{\theta \in \Theta} (t_r - \theta), \quad R_r = \frac{F_r}{A_r^5 \delta_r^6},$$
$$\Theta = \{0, -1/2, -1, 1/2399, -32/407\}.$$
$$W_r = \prod_{j=1}^{16} \left(1 + R_r \left(\frac{D_j}{\delta_r^{m_j+1} R_r^{m_j}} \right)^2 \right), \qquad b_r = \frac{R_r}{\mathrm{Tr}(W_r^{-1})^2}.$$

All displayed denominators and the inherited guards are retained. Each b_r receives a pole certificate.

Each first pole conic is covered by two charts, with finite bit $\varepsilon \in \{0, 1\}$ and retained parameters τ, s:

$$e = \frac{\tau^2 + (4\varepsilon - 1)B}{\tau^2 + 3B}, \qquad p = \frac{-4B\tau(1-\varepsilon)}{\tau^2 + 3B}, \qquad \tau^2 + 3B \neq 0.$$

For $e \neq 1$ choose $\tau = p/(e-1)$, $\varepsilon = 0$; the exceptional point $e = 1, p = 0$ uses $\varepsilon = 1$, $\tau = 0$. The remaining q is a finite quadratic root. Zero roots are included by the finite-algebra zero/nonzero case split.

3.2 A linear scalar normalization

The fixed scalar normalization is

$$\ell = 76249, \quad r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1),$$
$$R_L = \frac{102^2 \ell^L r}{(1 + C_L r)^2}, \quad C_L = 17(\ell^{(L+1)/2} + 1). \tag{5}$$

Here $L \geq 3$ is odd. The scalar norm tree and its six rational component tests are unchanged: they concern $f(H_0), f(S - P), z(S - P)/z(H_0)$ with orders $1, 1, 1, 1, 20, 20$. Its output W supplies the scalar pole input $R_L W^2$.

The finite residue certificate verifies that the five contact values are not attained by $216x^4 + 450459$ modulo ℓ, and that $1 + 17r$ never vanishes there. Consequently, for every rational input,

$$v_\ell(R_L) = \begin{cases} L, & v_\ell(X) \geq 0, \\ L - 20v_\ell(X), & v_\ell(X) < 0. \end{cases} \tag{6}$$

These valuations are positive and odd. At a consumed contact prime different from ℓ, the denominator is a unit and the contact valuation is unchanged. For integral X, the reduced denominator of R_L is square, it has the required local squareclasses and depths at $2, 3, 17$, and

$$0 < R_L < \frac{36}{\ell r} < 1.$$

The scalar norm argument therefore gives

$$v_\ell(a-1) \geq 19L + 1. \tag{7}$$

The normalization calculation in [1], together with the finite-field avoidance certificate [4], proves these statements and the preparation-zero realization. The supplied checker independently verifies every residue modulo 76249 and the normalization identity as an exact polynomial identity.

4 Soundness

The affine guard for $S - P$ excludes $a = 1$. Let $c = |a - 1| > 0$ and $B = \max(|n|, |a|, |t|, |b|)$. Equation (7) and $L \geq 3$ give $c \geq \ell^{19L+1}$ and therefore $B \geq c - 1 > 4$. In particular, the elementary residual bound $|\Delta| \leq 5B^4$ used below is valid. If the two contact radicals at the scales (4) have their kth powers dividing $N \neq 0$, the contact-height theorem gives

$$h(\lambda),\ h(A_1\lambda) \leq (H + \log 3)|N|^{19/k}.$$

Since $h(A_1) = 2h(\operatorname{Tr} \sum_i x(Q_i))$, the height inequality for a quotient and (1) give constants $K \geq 1$, $J \geq 0$, independent of X, L, such that

$$B^2 \leq K|N|^{19/k} + J. \tag{8}$$

The local formal-logarithm arguments of the source are unchanged; hence, with $\Delta = t^2 - a(a-2)tb + (a-2)^2b^2 - 1$, they imply

$$h(\beta) \leq Hc^{19/20}, \tag{9}$$

$$\Delta \neq 0 \Longrightarrow B^2 \leq K(5B^4)^{19/39} + J, \tag{10}$$

$$\beta \neq n \Longrightarrow B^2 \leq K(2e^{h(\beta)}B)^{19/10} + J. \tag{11}$$

Choose a fixed threshold T greater than each of

$$\begin{gathered} 16, \qquad 2\sqrt{2J} + 2, \qquad 2(2K5^{19/39})^{39/2} + 2, \\ (128H)^{20}, \qquad 8\big(10\log(2K) + 19\log 2\big). \end{gathered} \tag{12}$$

For $c > T$, one has $B \geq c - 1 > c/2$ and $B^2 > 2J$. If $\Delta \neq 0$, equation (10) gives $B^{2/39} \leq 2K5^{19/39}$, contrary to (12). Thus $\Delta = 0$. The elementary Lucas growth lemma proved in [2] then gives $\log B > c - 4$. If $\beta \neq n$, equation (11) gives

$$\log B \leq 10\log(2K) + 19\log 2 + 19h(\beta) < \frac{c}{8} + \frac{19c}{128} < c - 4,$$

using (9) and the threshold. This contradiction proves $\beta = n \in \mathbb{Z}$. Finally the elementary input-map lemma of [2] says that $216X^4 + 450459 \in \mathbb{Z}$ for rational X if and only if $X \in \mathbb{Z}$.

By (7), every source solution has $c \geq \ell^{19L+1}$. Therefore every odd L with $\ell^{19L+1} > T$ satisfies the soundness conclusion. The constant T is fixed but ineffective because J is ineffective.

5 Completeness

Fix an integer input X and a fixed odd L. All numerator and denominator primes of (5) form a finite set. The source realization first chooses the odd root labels with the scalar label $a-1$ sufficiently divisible and deep at these primes, and with the corresponding points in the required real neighborhoods. The explicit Lucas construction then gives odd nonzero t, b with $\Delta = 0$ and arbitrarily large growth. The input label is chosen to be $n = \beta$.

At this stage the two scales in (4) are fixed positive rational squares. Apply the same simultaneous prime-pattern theorem to these two scales. It chooses the slope and contact primes with the prescribed conjugate reduction orders while avoiding the previously fixed odd modulus. The subsequent Chinese remainder choice of the auxiliary index h and the points $U_i = [hn_i]P$ supplies the sixteen local tests to their assigned precisions and preserves the fixed scalar requirements and real neighborhoods. This is precisely the simultaneous construction of [3, 2]; its prime-pattern input accepts any fixed finite collection of the positive square scales and any fixed finite avoidance modulus.

Each of the two comparison inputs is thereby prepared for the zero-preparation pole theorem. The scalar input is prepared as well: R_L has square reduced denominator and the required local squareclasses, the six normalized scalar tests can be made arbitrarily small at the fixed finite places, and the scalar norm output can be made arbitrarily close to 1 in the real absolute value. Thus $0 < R_L W^2 < 17$, with the requisite denominator and local conditions. The three pole certificates exist by the inherited completeness input. The direct point chart (3) covers all nine chosen points; its multiplier needs no further choice. The two conic charts cover every pole certificate, and the zero/nonzero quadratic split includes vanishing second roots. All inherited guards are maintained by the same simultaneous realization.

This proves completeness for every fixed odd $L \geq 3$. Together with the preceding section it proves the arithmetic assertion for every sufficiently large such L.

6 Elimination of finite roots and the uniform degree

The retained coordinates and finite roots are as follows.

Retained rational coordinates	Count
Input	1
Nine square-root-abscissa pairs	18
Comparison slope	1
Two comparison-pole parameter pairs	4
Scalar-pole parameter pair	2
Total before finite-algebra elimination	26

There are nine quartic point roots, three quadratic pole roots, and three finite conic-chart bits. The maximum finite-algebra rank is $4^9 2^3 = 2^{21}$; the chart choices have $2^3 = 8$ branches. The guarded finite-algebra theorem of [2] replaces all twelve roots by one rational coordinate, including all zero-root branches and every nonzero denominator condition. This gives twenty-seven total polynomial variables.

The compiler specifies the polynomial by multiplication matrices in each finite algebra. It forms $H(V) = \det(I - VM_{G\lambda})$, the adjugate traces for the closed residuals, the leading coefficient and derivative resultant of each degree case, and the reciprocal polynomial excluding zero guards. A sum of squares imposes each branch's equalities over $\mathbb{Q}$, and a finite product takes the union of all chart, quadratic-support, separating-element, and degree cases. The exact finite index ranges and integer coefficient DAG are supplied with the computation.

The independently checked degree majorant is

$$D_* = 232075682770882905150152990100588354340946706432 < 2^{158}.$$

It is uniform in L. The uniform-degree checker replaces ℓ^L and $\ell^{(L+1)/2}$ by independent coefficient symbols, each of degree zero in the twenty-seven displayed coordinates. It makes no substitution for these symbols during the degree calculation. Integer specialization cannot increase the degree. The monic generator weights are

$$6, 6, 6, 6, 6, 6, 6, 6, 6, 1568138, 1935231, 116318.$$

The maximum basis weight is 3619849, the guarded multiplication-operator entry degree is 9979089, and the exact number of factors is 630360244960710623232. The independent recurrence for the determinant, adjugate, resultant, reciprocal, and final product gives D_*. It checks degrees only; no coefficient-norm bound for the unspecified sufficiently large L is inferred.

7 Reproduction and the status of a numerical example

The directory `trace27_cubic_family` contains the source builder, finite-algebra compiler, independent source and polynomial verifiers, and a separate uniform-degree verifier. The command

```
python3 reproduce.py --anchor-exponent 137731
```

reconstructs and checks a concrete family member. The exponent is a coefficient choice; changing it never adds a quantified variable. The packaged numerical member has a 7369-node source DAG and a 3032-node compiled integer coefficient DAG. The independent point checker expands every direct-chart coordinate identity over $\mathbb{Z}[h_0, h_1, Y]$. The polynomial verifier checks the nine quartics again, three rational-square clearings, all quadratic support cases, and 101 atomic nonzero guard factors. The linear normalizer is checked at every one of the 76249 residue classes.

For that numerical member the coefficient arithmetic and the bound $2^{2^{168}}$ are verified. This is a bound for the specified algebraic example. It is not a coefficient-height bound for a polynomial proved here to define $\mathbb{Z}$: no argument in this paper certifies that $L = 137731$ exceeds the ineffective threshold. The proved optimization is the twenty-seven-variable existence theorem with its uniform degree bound.

References

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[2] *An explicit polynomial in 29 variables over the rationals*, the established 29-variable source and guarded finite-algebra construction, version of 8 November 2026, `few_variables_20261108_v8.tex`. https://github.com/JoseBrox/diophantine-z-q_improvements.

[3] *Arithmetic dependencies of the construction with 29 variables*, version of 8 November 2026, `arithmetic_dependencies_20261108_v8.tex`. https://github.com/JoseBrox/diophantine-z-q_improvements.

[4] *Scalar contact normalization by finite-field avoidance*, companion scalar-normalization paper, 8 November 2026.

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[7] J. von Buhren, *Borne de hauteur semi-effective pour le problème de Mordell–Lang dans une variété abélienne*, J. Théor. Nombres Bordeaux **29** (2017), no. 1, 307–320, Theorem 1.1. https://www.numdam.org/item/10.5802/jtnb.981.pdf.